Yernur Melsov

AI Engineer · LLM, RAG & Multi-Agent Systems

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Flexible hours: full EU overlap by default, can work US hours. B2B contractor or via EOR (Deel / Remote.com).

Summary

AI engineer with 2+ years building production LLM systems, on a full-stack engineering foundation going back to 2022. Designs and ships **RAG pipelines** (GraphRAG on Neo4j, RAPTOR, ColBERT reranking) and **multi-agent systems** end-to-end, from data ingestion through evaluation to APIs and UI. Core stack: Python and TypeScript.

Experience

KOZ AI Jan 2025 – Present

AI Engineer · AI audits and custom AI integrations for business

- Shipped to production a **GraphRAG system** that checks and drafts compliance documents for a 300-person pharmaceutical client over **∼3,500** documents (Neo4j knowledge graph + PostgreSQL/pgvector, Gemini embeddings, ColBERT reranking, hierarchical RAPTOR indexing). **85%** of queries are resolved without escalation to a human expert.
- Cut response time **from 5 minutes to 1 minute** with a multi-agent system (dynamic query routing, end-to-end tracing), which turned a batch tool into an interactive assistant now used in live client calls and, internally, as our project-management agent.
- Built the ingestion pipeline with ontology-guided LLM entity and relation extraction: coreference resolution, entity deduplication, Leiden community detection.
- Designed retrieval as HyDE + **5 parallel retrievers** (vector, graph, ColBERT, RAPTOR, DRIFT) with map-reduce synthesis to cover the multi-hop and corpus-wide aggregation queries that single-vector search misses; answers are citation-grounded and pass agentic validation.
- Serve **∼3,000 requests/day** with a document auto-fill agent in Mastra (TypeScript) for **25 operators across 3 departments**: cut per-document preparation **from ∼25 to ∼5 minutes** across thousands of form types while keeping the document rejection rate at zero.
- Gate every pipeline change on evals: **RAGAS** faithfulness and context-recall on a golden query set, traced in **LangSmith** and checked with **DeepEval**, so regressions are caught before release.

Pleep Sep – Oct 2024

Full-Stack Developer (Contract)

- Improved a production agent-based client onboarding system: agent scenarios and UI (React/TypeScript), streaming agent responses, progress autosave between steps.

nFactorial Incubator Jun – Aug 2024

Full-Stack Developer (Startup Incubator Program)

- Built an AI platform for national university entrance exam preparation (OpenAI API + LangChain): **1,500 users** over one summer, and the project received a **$10,000 OpenAI API grant**.

Freelance Sep 2022 – May 2024

Full-Stack Developer

- Delivered **10+ turnkey projects** (React/Next.js, Express, PostgreSQL/MongoDB) for international clients: requirements gathering, development, deployment, and support, all async and remote across time zones.

Skills

RAG & Retrieval: GraphRAG (Neo4j), RAPTOR, HyDE, agentic RAG, semantic chunking; ColBERT reranking; pgvector, Qdrant, Pinecone, Weaviate

LLM & Agents: Google ADK, Mastra, LangChain, multi-agent orchestration, dynamic routing, tool calling, structured outputs; OpenAI / Gemini API

Evaluation & Observability: RAGAS, LangSmith, DeepEval, golden sets, LLM-as-judge, end-to-end tracing

Programming: Python, TypeScript/JavaScript

Backend & Data: FastAPI, Express, Next.js, React; PostgreSQL, MongoDB, Redis

Infra: Docker, GitHub Actions, Linux

Education & Languages

SDU University, Almaty · BSc in Information Systems, 2021–2025

Kazakh (native) · Russian (fluent) · English (professional working proficiency)